

1. General

moveEZ is an R package that animates PCA biplots, constructing biplots with `biplotEZ` and animating them with `gganimate`. The key idea is that some data has a time dimension, and so one might want to animate a PCA biplot over time. moveEZ exposes three plotting functions and one diagnostic function:

- `moveplot()`: embed data from all time points into the same shared subspace,
- `moveplot2()`: embed data from all time points into distinct subspaces of the same dimension, leaving the user to align them via the `align.time` and `reflect` arguments,
- `moveplot3()`: embed data from all time points into distinct subspaces of the same dimension, and then align them automatically by generalized Procrustes analysis (GPA), and
- `evaluation()`: report Procrustes-based measures comparing each aligned time slice against the target configuration.

The paper is clearly written and easy to follow. The package installs cleanly and the worked examples run.

2. Comments

1. The data examples in the paper demonstrate the functionality of the code, but they do not strongly motivate the code. What should a reader take away from Figures 7 and 8, from a substantive point of view? The manuscript would be much stronger if it invested in explaining the value of these visualizations to the reader, rather than leaving the reader to interpret the figures on their own. I personally struggled to find substantive takeaways from the figures.
2. The manuscript lacks a formal mathematical description of Generalized Procrustes Analysis, which is used to produce the figures from `moveplot3()`. This makes it difficult to understand what is actually being visualized.
3. It is reasonable to want to align various principal component analyses. However, the manuscript takes as given that these various decompositions should be aligned relative to a consensus GPA alignment. Is there any particular reason for this? Does the consensus GPA alignment preserve the principal component analysis interpretation of the axes, or is this a new kind of object that must be interpreted differently (I suspect it is the latter)? Methodologically, it could make more sense to use a tensor decomposition method that embeds the various slices into a directly interpretable subspace. Alternatively, if the embedding subspace is immaterial, generalized Procrustes analysis is computationally unnecessary and a single alignment step to any target is sufficient for comparability.
4. In `evaluation`, `MB` is definitionally zero due to a missing `abs()`. Further, in `evaluation()` `X.new` is never used. `AMB`, `MB`, and `RMSB` are all computed from `target - testee`, where `testee` is the un-rotated, un-scaled configuration. Is this intended? `PS`, by contrast, does use the rotated configuration.

3. Code

The code is readable and well organised. Some comments:

1. The names `moveplot()`, `moveplot2()`, and `moveplot3()` are not informative about what the functions do.
2. When given anything other than a `moveplot3` object, `evaluation()` calls `print()`. `stop()` would be the conventional choice here.
3. In `evaluation()`, `Tot.SS`, `Fit.SS`, `n.X`, `p.X`, and `p.Y` are computed but never used, and `X.new` is computed but never used. Is this intended?

4. Typos

- pg. 8: "These aspects are are addressed in `moveEZ`."
- pg. 11: "the squared sum of squares distances"

5. Summary

My view is that the contribution is not substantial in the sense the R Journal requires. The alignment step is a well-known technique that is straightforward to implement (see below) and once the data has been aligned, animating the biplots is a straightforward combination of `ggplot2` and `gganimate`.

```
library(gapminder)
library(vegan)

pca_vars <- c("lifeExp", "gdpPercap", "pop")
years    <- sort(unique(gapminder$year))

# scale once globally
scaled_vars <- scale(gapminder[pca_vars])

# one PCA score matrix per year, aligned by country so that row i always
# refers to the same country -- GPA/Procrustes requires the rows of each
# configuration to correspond to the same entities
configs <- lapply(years, function(yr) {
  rows      <- gapminder$year == yr
  by_country <- order(gapminder$country[rows])
  prcomp(scaled_vars[rows, ][by_country, ])$x
})

# --- generalized Procrustes analysis -----
# Start from one year as an initial guess at the consensus shape, then
# iterate: align every year to the current consensus, recompute the
# consensus as the mean of the aligned configurations, and repeat until
# the consensus stops moving.

consensus <- configs[[1]]

repeat {
  aligned <- lapply(configs, function(cfg) procrustes(X = consensus, Y = cfg)$Yrot)

  new_consensus <- Reduce(`+`, aligned) / length(aligned)
  shift <- sqrt(sum((new_consensus - consensus)^2))
```

```
consensus <- new_consensus

  if (shift < 1e-8) break
}

aligned_by_year <- setNames(aligned, years)
```

It may be true that this functionality has not been cleanly wrapped into a single function before, but the convenience of a wrapper around this functionality is quite modest. Further, the manuscript does not justify its methodological choices in terms of any properties of generalized Procrustes analysis or common variance structures or shared subspaces, so the interpretation of the figures produced by moveEZ is ambiguous.